Contents

1 A sequence from early West Germanic consonant and vowel shifts to Old English r-metathesis

1.1 Introduction

The sequence begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

1.2 Numbering note

The sequence follows the established rule numbering.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC077 is unused.

2 West Germanic rhotacism

2.1 Historical discussion

Hogg states that Germanic *z* yielded *r* in intervocalic position in Old English, while final *z* was generally lost (Hogg 1992, 37). Ringe and Taylor argue that this merger of *z* with *r* was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic *z*-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

The label SC003 PGmcRhotacism is historically misleading: the change is a later West Germanic rhotacism, not a Proto-Germanic one. It is also distinct from SC020 PGmcFinalZDeletion, which removes final *z* before the surviving medial consonant becomes *r*.

2.2 SC003. West Germanic rhotacism (`PGmcRhotacism`)

```
define PGmcRhotacism [
```

```
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc **líznōjaną* yields *lirnian* rather than expected OE *liornian* 'learn', PGmc **líznōθi* yields *lirnaþ* rather than expected *liornaþ*, PGmc **líznô* yields *lirna* rather than expected *liorna*, and PGmc **mízdai* yields *merde* rather than expected OE *meorde* 'meed'. Moving rhotacism earlier within the tested range changes none of the checked forms.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. Its placement after the earlier loss of final **z* rests on the historical analyses cited above.

3 Proto-West-Germanic ai-monophthongization

3.1 Historical discussion

Ringe and Taylor treat the reduction of unstressed **ai* as one of the major early vowel shifts shared across the Northwest Germanic area (Ringe and Taylor 2014, 40–41).

The historical support is strongest for unstressed **ai*, especially word-finally. The rule extends the change to nonfinal **ai* > **ā*, a generalization stated more sharply than in the current handbook discussion. Both developments have inherited **ai* as their input.

3.2 SC004. Proto-West-Germanic ai-monophthongization (PWGmcAiMonophthongization)

```
define PWGmcAiMonophthongization [
    [{*ai} -> {*ē} || _ .#.]
    .o.
    [{*ai} -> {*ā}]
    .o.
    [{*ái} -> {*ā}]
];
```

The soul form fixes the relation to interstress raising. If monophthongization is delayed until after that change, PGmc **sáiwalō* yields *sāwel* rather than expected OE *sāwol* 'soul'. No earlier placement changes a checked output.

This witness proves that monophthongization preceded interstress raising; it says nothing about the date of the wider nonfinal **ai* > **ā* generalization. The word-final merger with long mid **ē* belongs among the early Northwest Germanic vowel shifts; the broader chronology remains less certain.

4 Unstressed **a*-raising before final **m*

4.1 Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

4.2 SC005. Unstressed **a*-raising before final **m* (`NWGmcAToUBeforeM`)

```
define NWGmcAToUBeforeM [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
     ↪  ({*z})? .#.
];
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 NWGmcULowering, PGmc **skúldramiz* yields *sċoldrum* rather than expected OE *sċuldrum*; earlier placements converge on the expected output. The `shoulder` family therefore tests the chronology, while the inflectional endings justify restricting the change to noninitial unstressed material before **m*.

The evidence is confined to inflectional material.

5 Early i-apocope

5.1 Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geoguþ* 'youth' (Brunner 1965, sects 145–146; Ringe and Taylor 2014, 141). Campbell's *dugup* and *geogup* examples belong to the same pattern (Campbell 1959, sect. 332).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

5.2 SC006. Early i-apocope (`PWGmcEarlyIApocope`)

```
define PWGmcEarlyIApocope [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
];
```

The absence of umlaut in *geoguþ* 'youth' provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLong-Diphthong, PGmc **skáwōθi* yields *sċēaweþ* rather than expected OE *sċēawaþ*.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every checked output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

6 Final *ō-lowering before *r

6.1 Historical discussion

Ringe and Taylor treat the West Germanic lowering of final bimoric **ō* before word-final **r* as a specific inherited development and illustrate it above all with the families behind *fēower* 'four' and *wæter* 'water' (Ringe and Taylor 2014, 58–59).

The rule is historically secure but narrow: final or pre-final **ō* before word-final **r*. The clearest evidence remains concentrated in the `four` and `water` material. No broader environment for **ō* is attested.

6.2 SC007. Lowering of final bimoric *ō before *r (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering [
    {*ō} -> {*a} || _ {*r} .#.
];
```

OE *wæter* 'water' reveals why lowering must precede SC043 AngloFrisianBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* yields *water* rather than expected OE *wæter* 'water': brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no checked output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. The *fēower* 'four' and *wæter* 'water' families support the narrow environment before word-final **r*; no broader lowering of **ō* is attested.

7 Coronal-w assimilation

7.1 Historical discussion

Ringe and Taylor treat the assimilation of `*dw` and `*zw` to `*ww` as a shared Proto-West-Germanic innovation and support it through the `four` family and plural-pronominal forms such as `you` and `your` (Ringe and Taylor 2014, 56–57).

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but only the numeral and the pronominal forms directly support the generalization.

7.2 SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
];
```

OE *fēower* 'four' exposes a feeding relation: coronal assimilation must create **ww* while simplification can still reduce it. If SC008 PWGmcCoronalWAssimilation is delayed until after SC031 OEWWSimplification, PGmc **fédwōr* yields *fēowwer* rather than expected OE *fēower* 'four'. Earlier placements alter no checked output.

The numeral fixes that relative order. The pronouns extend both input clusters beyond `four`; the earlier boundary remains undetermined.

8 *ij*-contraction in *friend*

8.1 Historical discussion

Ringe and Taylor describe a change of `*ijo` to `*iu` in the `friend` family, with the pathway PGmc **frijond-* > PWGmc **friund* > OE *frēond* 'friend' (Ringe and Taylor 2014, 62). The same source immediately warns that the `*ijo` sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62).

The change concerns a rare sequence confined to the `friend` family and cannot safely be generalized into a broadly productive rule.

8.2 SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

```
define PWGmcIjContraction [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

Only the `friend` family tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* yields *friund* rather than expected OE *frēond* 'friend'; moving contraction earlier within the tested range changes no checked output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond this family, precisely the reservation made by Ringe and Taylor.

9 West Germanic j-gemination

9.1 Historical discussion

Fulk treats West Germanic consonant gemination before `*j` after a short vowel as a regular development and illustrates it with forms such as OE *settan* ‘set’ and *lecgan* ‘lay’ (Fulk 2018, 127, §6.15).

The change applies specifically after a short vowel before **j*, not to geminate consonants generally.

9.2 SC010. West Germanic j-gemination (`PWGmcJGemination`)

```
define PWGmcJGemination [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

OE *nett* ‘net’ fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátją* yields *nete* rather than expected OE *nett* ‘net’. Earlier movement of gemination changes no checked output.

The chronology is phonologically transparent: the consonant must geminate before **j* ceases to be consonantal. The witness establishes no earlier boundary.

10 Syllabic j after final-vowel loss

10.1 Historical discussion

Ringe and Taylor state directly that after final unstressed `*a` and `*ą` were lost, postconsonantal `*j` became syllabic `*i`, with outcomes behind OE *here* 'army' and *rice* 'kingdom' (Ringe and Taylor 2014, 46).

The sources establish the development, although the checked lexicon supplies little independent evidence for its position. Its scope is postconsonantal j, not high-vowel vocalization generally.

10.2 SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

```
define PWGmcSyllabicJ [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

The same PGmc **nátją* witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGemination yields *nete* rather than expected OE *nett* 'net'; moving it later changes no checked output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* and *rice* as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

11 *lþ*-voicing

11.1 Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan*, *beald*, *wuldor*, and *gylden* (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold*, and *feld* (Campbell 1959, 169, §414).

The comparative evidence supports *lþ* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

11.2 SC012. *lþ*-voicing (`PWGmcLThVoicing`)

```
define PWGmcLThVoicing [
    {*θ} -> {*d} || {*l} _
];
```

The `field`, `fold`, `gold`, and `wold` families preserve **lþ* to **ld*, but none dates the change against a neighboring rule. Every checked output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lþ* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

12 Dental hardening

12.1 Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative *ð became stop *d in all positions (Ringe and Taylor 2014, 43).

The change is systemic across early West Germanic and extends beyond any one lexical family.

12.2 SC013. Dental hardening (`PWGmcDentalHardening`)

```
define PWGmcDentalHardening [
    {*ð} -> {*d}
];
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no checked output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

13 Early unstressed vowel changes

13.1 Historical discussion of the earliest unstressed vowel changes

The first change removes the remaining diphthongal quality of unstressed **ai*; the second carries early unstressed front-vowel leveling farther in forms such as *weorold* 'world'. Their chronological evidence differs: monophthongization is historically clear but not closely dated by the witness forms, whereas **i*-lowering has a diagnostic later boundary.

13.2 Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the broad Northwest Germanic reduction of unstressed **ai* to a long mid vowel that merges with unstressed **e* (Ringe and Taylor 2014, 37–41). The historical change is thus established, although the order test determines no closer relative position.

13.3 SC014. Monophthongization of unstressed **ai* (NWGmcUnstressedAiMonophthongization)

```
define NWGmcUnstressedAiMonophthongization [
    {*ăi} -> {*ē}
];
```

Moving SC014 NWGmcUnstressedAiMonophthongization earlier or later changes no checked form. The lexicon therefore cannot refine its source-based placement among the earliest Northwest Germanic simplifications of unstressed vowels.

Ringe and Taylor's merger of unstressed **ai* with **e* establishes the historical development; the current witnesses do not distinguish its position relative to neighboring changes.

13.4 Historical discussion of early unstressed front-vowel leveling

Campbell treats the merger of unstressed front vowels directly and also records the variation of *weorold* and *weoruld* (Campbell 1959, 141–42, 154–55). These forms supply SC015 NWGmcILowering with a firmer lexical basis than the preceding change.

13.5 SC015. Leveling of early unstressed front vowels (`NWGmcILowering`)

```
define NWGmcILowering [
    {*i} -> {*e}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel,
    {*í} -> {*é}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel
];
```

The *weorold* and *weoruld* variants turn the general source claim into an ordering test. If SC015 NWGmcILowering is delayed until after SC036 OEInterStressRaising, PGmc **wír-àldu* yields *wuruld* rather than expected OE *weorold* 'world'; earlier movement changes no checked output.

The derivation thus fixes front-vowel leveling before interstress raising while leaving its earlier boundary open.

SC016 OEWsPalatalGlide and SC017 NWGmcULowering follow with a more tightly constrained local chronology.

14 West Saxon palatal glide and u-lowering

14.1 Historical discussion of West Saxon palatal glide and u-lowering

The derivation of *ġeoc* 'yoke' passes through both rules. Campbell treats the West Saxon rising-diphthong spellings before back vowels, while the same handbook tradition describes the lowering of *u* before a following non-high vowel separately (Campbell 1959, 17, §44; Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3).

The first change creates the West Saxon *ġeoc* type; the second carries the same material into the subsequent vowel history.

14.2 Historical discussion of West Saxon palatal glide

West Saxon spellings such as *ġeoc* 'yoke', *ġeong* 'young', and *ġeoguþ* 'youth' reflect an early development before back vowels. Campbell gives the most direct handbook statement of the phenomenon (Campbell 1959, 17, §44).

The sources establish SC016 OEWsPalatalGlide, although the checked forms provide only a later boundary.

14.3 SC016. West Saxon palatal glide before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
    {*j} {*u} -> {*j} {*e} {*u} || .#. _,
    {*j} {*ú} -> {*j} {*é} {*u} || .#. _
] .o. [
    {*ʤ} {*u} -> {*ʤ} {*e} {*u} || .#. _,
    {*ʤ} {*ú} -> {*ʤ} {*é} {*u} || .#. _
] .o. [
    {*ʧ} {*u} -> {*ʧ} {*e} {*u} || .#. _,
    {*ʧ} {*ú} -> {*ʧ} {*é} {*u} || .#. _
] .o. [
    {*ʃ} {*u} -> {*ʃ} {*e} {*u} || .#. _,
    {*ʃ} {*ú} -> {*ʃ} {*é} {*u} || .#. _
];
```

OE *ġeoc* 'yoke' fixes the close relation between glide insertion before back-vocalic *u* and the following change.

If glide insertion follows SC017 NWGmcULowering, PGmc **júką* yields *ġoc* rather than expected OE *ġeoc* 'yoke'; earlier placement changes no checked output. The witness therefore dates SC016 OEWsPalatalGlide before u-lowering without supplying an earlier boundary. The *ġeoc*, *ġeong*, and *ġeoguþ* material establishes the lexical scope of the West Saxon development.

14.4 Historical discussion of u-lowering

After the glide-conditioned West Saxon spellings are in place, the broader Northwest Germanic lowering of *u* to *o* before a following non-high vowel provides the clearest standard sound change in this small region. Campbell and Fulk both describe that change directly (Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3).

SC017 NWGmcULowering thus rests on a broader source base than the preceding West Saxon rule.

14.5 SC017. Lowering of **u* before following non-high vowels (NWGmcULowering)

```
define NWGmcULowering [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
];
```

Lowering of *u* to *o* is fixed on both sides by *ġeoc* ‘yoke’, *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’.

Before SC016 OEWsPalatalGlide, PGmc **júką* yields *ġoc* rather than expected OE *ġeoc* ‘yoke’. After SC019 NWGmcFinalLongORaising, PGmc **núsō* yields *nusu* rather than expected *nosu* ‘nose’, PGmc **skúflō* yields *sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* yields *surg* rather than expected *sorg* ‘sorrow’. The two witness sets place SC017 NWGmcULowering after glide formation and before final long-*o* raising.

15 Stressed monosyllable **ō*-raising

15.1 Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* ‘cow’, *hū* ‘how’, *tū* ‘two’, and *bū* ‘both’ (Campbell 1959, 47, §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

15.2 SC018. Raising of final stressed monosyllabic **ō* (NWGmcStressedMonosyllableORaising)

```
define NWGmcStressedMonosyllableORaising [
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
];
```

Campbell’s *cū*, *hū*, and *tū* establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 NWGmcStressedMonosyllableORaising with neighboring changes leaves every checked output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

16 Final long-*o* raising and final *z*-deletion

16.1 Historical discussion of final long-*o* raising and final *z*-deletion

The same final-syllable structure undergoes both changes. Ringe and Taylor describe the change of unstressed final non-nasalized long **ō* to short **u*, while Hogg and Crist treat word-final *z loss as a separate later step in West Germanic (Ringe and Taylor 2014, 30; Hogg 1992, 37; Crist 2002, 1).

The derivation of *ræste* ‘rest’ fixes their order: SC019 NWGmcFinalLongORaising must still see final **ō*, and SC020 PGmcFinalZDeletion removes final *z only afterward.

16.2 Historical discussion of final long-*o* raising

The first change in the pair is the Northwest Germanic raising of unstressed final long **ō* to **u*. Ringe and Taylor state that development directly in comparative terms (Ringe and Taylor 2014, 30).

The change supplies the final vowel of forms such as *nosu*, *sċofl*, and *sorg*.

16.3 SC019. Raising of final unstressed long **ō* (NWGmcFinalLongORaising)

```
define NWGmcFinalLongORaising [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
];
```

Two groups of witnesses confine final unstressed long **ō* > **u*. The forms *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’ fix its lower boundary.

Before SC017 NWGmcULowering, PGmc **núsō* yields *nusu* rather than expected OE *nosu* ‘nose’, PGmc **skúflō* yields *sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* yields *surg* rather than expected *sorg* ‘sorrow’. After SC020 PGmcFinalZDeletion, PGmc **rástōz* yields *rast* rather than expected *ræste* ‘rest’. These failures place SC019 NWGmcFinalLongORaising after u-lowering and before final *z*-loss.

16.4 Historical discussion of final *z*-deletion

The second change is the loss of word-final *z. Standard handbook tradition and Crist’s West Germanic discussion establish the development within broader accounts of inflectional morphology (Hogg 1992, 37; Crist 2002, 1).

Final z-loss follows long-o raising and precedes the later changes in weak syllables.

16.5 SC020. Deletion of word-final *z (`PGmcFinalZDeletion`)

```
define PGmcFinalZDeletion [{*z} -> 0 || _ .#.];
```

The chronology of word-final *z-loss is unusually well delimited: *ræste* supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 NWGmcFinalLongORaising, PGmc **rástōz* yields *rast* rather than expected OE *ræste* ‘rest’. After SC040 OEMedUnstressedULowering, PGmc **bébruz* yields *befro* rather than expected *befer* ‘beaver’, PGmc **kwéðuz* yields *cwedo* rather than expected *cwedu* ‘cud’, and PGmc **félθuz* yields *feldo* rather than expected *feld* ‘field’, alongside eight other newly failing rows. Final *z*-loss therefore follows SC019 NWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The **rástōz* derivation fixes the local relation to SC019 NWGmcFinalLongORaising. The distant boundary at SC040 OEMedUnstressedULowering shows only that word-final **z*-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final **ō*-raising.

17 Unstressed **o*-raising

17.1 Historical discussion

The older history of *heofon* 'heaven' requires an unstressed-vowel adjustment before the later reshaping of medial vowels in Old English. Campbell derives the *-o-* from an earlier unstressed environment, and Ringe and Taylor place the same family within the wider West Germanic record (Campbell 1959, 155–56, §373; Ringe and Taylor 2014, 287).

The change is historically recognizable, but the checked forms provide only a later boundary.

17.2 SC021. Raising of unstressed **o* before later **u* (NWGmcUnstressedORaising)

```
define NWGmcUnstressedORaising [
    {*o} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _
     ↪  EnglishStarConsonant* {*ų}
];
```

After SC040 OEMedUnstressedULowering, PGmc **xémonų* yields *heofun* rather than expected OE *heofon* 'heaven'; earlier placement changes no checked output. The witness therefore places SC021 NWGmcUnstressedORaising before medial unstressed-*u* lowering.

Nothing in the present lexicon supplies the corresponding earlier boundary.

18 *mn*-dissimilation

18.1 Historical discussion

The handbooks describe the history of *mn* sequences as a limited descriptive pattern. Campbell discusses both loss of unstressed material and later assimilation in forms of this type, including the special status of *month*-type evidence (Campbell 1959, 189, 195, §§470, 484).

The pattern is historically established, but the checked forms do not constrain its position.

18.2 SC022. Dissimilation of *mn* sequences (`NWGmcMnDissimilation`)

```
define NWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _
           EnglishStarVocalic EnglishStarConsonant* EnglishStarNasal
];
```

Campbell's *heofon* and *month* material supports early *m* > *β* before a later nasal, but supplies no ordering witness.

Moving SC022 NWGmcMnDissimilation earlier or later leaves every checked output unchanged. Its place among the early consonantal changes rests on the handbook account of *mn*-dissimilation.

19 N-stem *n*-loss

19.1 Historical discussion

The broader history is the reduction and leveling of older n-stem endings in West Germanic. Ringe and Taylor describe the resulting syncretism in the n-stems, which is the wider morphological setting for the narrower step isolated here (Ringe and Taylor 2014, 72).

The path to *dōn* 'do' provides the clearest witness, but the change remains narrow in scope.

19.2 SC023. Loss of n-stem **n* in final position (`NWGmcNStemNLoss`)

```
define NWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

After SC047 OEHeavySyllableNasalApocope, PGmc **dōną* fails entirely (+*?*) instead of yielding expected OE *dōn* 'do'; earlier placement changes no checked output. Thus SC023 NWGmcNStemNLoss must feed the later apocope.

This failed derivation supplies a terminus ante quem, while the lower boundary remains unattested.

20 Long *ē*-lowering

20.1 Historical discussion

The later West Saxon forms *sċēap* ‘sheep’ and *ġēar* ‘year’ imply an earlier lowering of long *ē* before the palatal diphthongal outcomes described more fully later in the sequence. Campbell and Ringe and Taylor discuss those later West Saxon outputs directly (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1).

The change is historically recognizable, but the checked forms provide only a later boundary.

20.2 SC024. Lowering of long *ē* before non-nasal consonants (NWGmcLongELowering)

```
define NWGmcLongELowering [
    {*ē} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal],
    {*ḗ} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

After SC056 OEWsPalatalDiphthongization, long *ē* > *ǣ* can no longer produce the expected West Saxon forms: PGmc **skḗpą* yields *sċīep* rather than OE *sċēap* ‘sheep’, and PGmc **jḗrą* yields *ġīer* rather than *ġēar* ‘year’. Earlier placement changes no checked output, so SC024 NWGmcLongELowering has a secure upper boundary.

Its lower boundary remains a matter of handbook chronology.

21 Long *ē* nasal-rounding

21.1 Historical discussion

Before nasals, older long *ē* can round toward the *ō*-vocalism seen later in *mōnaþ* 'month' and *mōna* / *mōn*-type material. Campbell treats this split directly in his discussion of Germanic long *ē* before nasal consonants (Campbell 1959, 53, §129).

The change is historically recognizable, but the tested forms supply no close relative chronology.

21.2 SC025. Rounding of long *ē* before nasals (`NWGmcLongENasalRounding`)

```
define NWGmcLongENasalRounding [
    {*ē} -> {*ō} || _ EnglishStarNasal,
    {*ḗ} -> {*ō} || _ EnglishStarNasal
];
```

Reversing SC025 NWGmcLongENasalRounding with neighboring changes leaves every checked output unchanged. Its position beside the other *ē*-developments therefore follows the handbooks.

22 Nasal spirant changes

22.1 Historical discussion of nasal loss before spirants and compensatory lengthening

The two rules state successive phases of a single development. Campbell describes nasal loss before voiceless spirants with compensatory lengthening and nasalization of the preceding vowel. Ringe and Taylor assign the same outcomes to inherited northern West Germanic, before late Old English (Campbell 1959, 47, §121; Ringe and Taylor 2014, 140–41).

SC026 NWGmcNasalSpirantLengthening adjusts the vowel while the nasal-plus-spirant sequence remains present; SC027 NWGmcNasalSpirantLoss then removes the nasal. The first rule must therefore precede the second.

22.2 SC026. Lengthening before nasal plus spirant (NWGmcNasalSpirantLengthening)

```
define NWGmcNasalSpirantLengthening [
    {*a} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*e} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*o} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*æ} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*á} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*é} -> {*ḗ} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ó} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative
];
```

All three witnesses require the vowel adjustment while the nasal is still present. If SC026 NWGmcNasalSpirantLengthening follows SC027 NWGmcNasalSpirantLoss, PGmc **fúnxstiz* yields *fyst* rather than expected OE *fȳst* 'fist', PGmc **gánsz* yields *ġeas* rather than expected *gōs* 'goose', and PGmc **júgunθ* yields *ġeogoþ* rather than expected *ġeoguþ* 'youth'. Earlier placement changes no checked output. The evidence requires lengthening to precede nasal loss without supplying a lower boundary, in agreement with the handbook treatment of the two as successive phases.

22.3 SC027. Loss of the nasal before spirants (`NWGmcNasalSpirantLoss`)

```
define NWGmcNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarVoicelessFricative
];
```

The converse test fixes the same boundary: placing SC027 NWGmcNasalSpirantLoss before SC026 NWGmcNasalSpirantLengthening produces the same errors in *fȳst* 'fist', *gōs* 'goose', and *ġeoguþ* 'youth'. Later placement changes no checked output. These forms prove that the vowel was adjusted before the nasal disappeared; they provide no upper boundary for the loss.

23 Preconsonantal **x*-loss

23.1 Historical discussion

Campbell explicitly treats loss of *x* and gives forms such as *fléam* 'flight' and *hēla* 'heel' as examples of the same broad development (Campbell 1959, 186, §461).

The historical evidence is firmer than the chronology: the checked forms do not constrain the rule's position.

23.2 SC028. Loss of preconsonantal **x* (`NWGmcPreconsonantalXLoss`)

```
define NWGmcPreconsonantalXLoss [
    {*x} -> 0 || _ {*s} EnglishStarConsonant
];
```

No witness word dates preconsonantal **x*-loss before **s* plus another consonant: moving SC028 NWGmcPreconsonantalXLoss in either direction leaves every checked output unchanged. Its position within this stretch therefore rests on the handbook chronology for *x*-loss.

24 Awj glide formation and au-fronting

24.1 Historical discussion of awj glide formation and au-fronting

The *hay* and *strew* material undergoes both changes. Glide formation reshapes the older *awj* sequence, and fronting then affects the resulting *au*. Campbell's discussion of these outcomes and Ringe and Taylor's derivations of *hīeġ* and *strīeġan* describe the same sequence (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

Glide formation creates the input to fronting; diphthong leveling follows both.

24.2 Historical discussion of awj glide formation

Older *awj* sequences are the source of forms such as *hīeġ* 'hay' and *strīeġan* 'strew'. Campbell treats the relevant developments directly, and Ringe and Taylor likewise trace the same material through intermediate *auj*-type stages (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

The sources establish glide formation, while the witness forms supply only a later boundary.

24.3 SC029. Glide formation in **awj* (`OEAwjGlideFormation`)

```
define OEAwjGlideFormation [
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j},
    {*á} {*w}      {*j} -> {*áu} {*j},
    {*a} {*w}      {*j} -> {*au} {*j}
];
```

The *hīeġ* and *strīeġan* derivations show that *awj* reshaping prepared the input to fronting. If fronting is applied first, PGmc **xáwwją* yields *hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* yields *strauian* rather than expected *strīeġan* 'strew'. Earlier placement of glide formation changes no checked output, so these forms supply an upper boundary without a corresponding lower one.

24.4 Historical discussion of au-fronting

Once the glide sequence is in place, *au*-fronting produces the fronted diphthongal outcomes of the broader West Saxon vowel history. Campbell describes *au* > *ēa* (Campbell 1959, 53–54, §135).

Fronting must follow glide formation and precede diphthong leveling, which applies to a wider set of derivations.

24.5 SC030. Fronting of **au* (`OEAuFronting`)

```
define OEAuFronting [
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
];
```

Two distinct failure sets confine fronting. Placed before glide formation, it produces the wrong forms: PGmc **xáwwją* yields *hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* yields *strauian* rather than expected *strīeġan* 'strew'. Placed after diphthong leveling, PGmc **galáubijaną*, **bráudą*, and **dráugmaz*, together with sixteen other derivations, fail to produce output at all (+*?*) instead of yielding expected OE *ġelīefan* 'believe', *brēad* 'bread', and *drēam* 'dream'. The lexical errors require fronting to follow glide formation, while the failed derivations require it to precede diphthong leveling.

The later failure set consists of failed derivations, not competing Old English surface forms.

25 West Saxon diphthong sequence

25.1 Historical discussion of the West Saxon diphthong sequence

Four distinct developments shape the West Saxon diphthongal field. Campbell discusses inherited *aw*/*ew* outcomes, palatal-triggered diphthongization, and later Anglian smoothing in connected but separate parts of the vowel history; Hogg likewise distinguishes the palatal-diphthongal developments (Campbell 1959, 46, 53–54, 65–70, 95–96, §§120, 135–136, 170–176, 185, 223–227; Hogg 1992, 106–7, 111–12).

The closest interaction joins *ww*-simplification and long-*aw* diphthongization, which together shape *dēaw* 'dew' and *hēawan* 'hew'. Diphthong leveling regularizes a wider field, while long-*ew* diphthongization carries *ēow* into the later environment of breaking.

25.2 Historical discussion of WW simplification

West Germanic *ww* sequences lie behind forms such as *dēaw* 'dew' and *hēawan* 'hew', and Campbell treats them as part of the early West Germanic diphthong history (Campbell 1959, 46, §120).

SC031 OEWWSimplification precedes the later long-diphthong outcomes.

25.3 SC031. Simplification of **ww* sequences (`OEWWSimplification`)

```
define OEWWSimplification [
    {*w} {*w} -> {*w}
];
```

The *dēaw* and *hēawan* derivations establish that doubled *w* was simplified before the long *ēaw* development. If SC031 OEWWSimplification follows SC034 OEAwLongDiphthong, PGmc **dáwwō* yields *dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* yields *hawan* rather than expected *hēawan* 'hew'. Earlier placement changes no checked output. The witnesses require simplification before the long-diphthong change and leave the lower boundary to the broader West Saxon chronology.

25.4 Historical discussion of diphthong leveling

Forms such as *hēafod* 'head' reflect the redistribution of diphthongal outcomes across a wider set of words. Campbell describes smoothing and related later monophthongization, although the rule below is more narrowly conditioned than any single textbook label (Campbell 1959, 95–96, §§223–227).

The evidence for SC032 OEDiphthongLeveling is less self-contained than that for the *dēaw* / *hēawan* developments.

25.5 SC032. Leveling of diphthongal outputs (`OEDiphthongLeveling`)

```
define OEDiphthongLeveling [
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
];
```

The two edges of this interval fail differently. Before SC030 OEAuFronting, PGmc **galáubijaną*, **báug*, and **bráudą* produce no output (+?) instead of expected OE *ġelīefan* 'believe', *bēag* 'bow', and *brēad* 'bread', alongside fifteen other failed derivations. After SC040 OEMedUnstressedU-Lowering, PGmc **xáubudą* yields *hēafud* rather than expected *hēafod* 'head'. Absence at the lower edge places diphthong leveling after fronting; the wrong surface form at the upper edge places it before medial unstressed-*u* lowering.

25.6 Historical discussion of long *ēow*

The long *ēow* forms of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' form part of the West Saxon vowel history, although their clearest ordering relation points forward. Campbell describes early *eu* in Old English, and Ringe and Taylor give the corresponding examples from *chew*, *four*, and *knee* (Campbell 1959, 53–54, §136; Ringe and Taylor 2014, 188, 202).

The only checked boundary for SC033 OEEwLongDiphthong lies ahead at SC044 OEBreaking.

25.7 SC033. Long *ēow* before following vowels and weak endings (OEEwLongDiphthong)

```
define OEEwLongDiphthong [
    {*e} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*i} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*é} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*í} {*w} -> {*ēo} {*w} || _ OEEwLongContext
];
```

The long *ēow* of *ċēowan*, *fēower*, and *cnēow* supplies only a terminus ante quem. If SC033 OEEwLongDiphthong follows SC044 OEBreaking, PGmc **kéwwaną* yields *ċeowan* rather than expected OE *ċēowan* 'chew', PGmc **fédwōr* yields *feower* rather than expected *fēower* 'four', and PGmc **knéwą* yields *cneow* rather than expected *cnēow* 'knee'. Earlier placement changes no checked output. The sources associate *ew* and *iw* with the same diphthongal history but furnish no lower boundary.

25.8 Historical discussion of long *ēaw*

After SC031 OEWWSimplification has reduced *ww* to single *w*, the remaining *aw* sequence can develop into the long *ēaw* seen in *dēaw* and *hēawan*. Campbell treats these outputs in the early diphthong history of West Germanic and Old English (Campbell 1959, 46, 53–54, §§120, 135–136). The resulting long diphthong is *ēaw*.

SC034 OEAwLongDiphthong follows SC031 OEWWSimplification locally and must also precede SC043 AngloFrisianBrightening.

25.9 SC034. Long *ēaw* before following vowels (`OEAwLongDiphthong`)

```
define OEAwLongDiphthong [
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    {*á} {*w} -> {*ḗa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

A local feeding relation and a later vowel change confine *aw* > *ēaw*. Before SC031 OEWWSimplification, PGmc **dáwwō* yields *dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* yields *hawan* rather than expected *hēawan* 'hew'. After SC043 AngloFrisianBrightening, PGmc **skáwōjaną* yields *sċawian* rather than expected OE *sċēawian* 'show', PGmc **skáwōθi* yields *sċawaþ* rather than expected *sċēawaþ*, and PGmc **stráwą* yields *stræw* rather than expected *strēaw* 'straw'. The *dēaw* and *hēawan* forms require long-diphthong formation after simplification, while *sċēawian* requires it before brightening; the handbooks assign the same interval to the West Saxon development.

26 Prefix and compound adjustments

26.1 Historical discussion of prefixal **a*-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* supplies the comparative witness for the same development (Campbell 1959, 147, §354; Ringe and Taylor 2014, 245; 2014, 267).

The rule has a narrow historical range and gives prefixed forms the weak vowel inherited by later vocalic changes.

26.2 SC035. Reduction of prefixal **a* (`OEPrefixAReduction`)

```
define OEPrefixAReduction [
    {*a} -> {*ĕ}
        || .#. {*g} _
           [EnglishStarConsonant | EnglishPalatalConsonant]
           EnglishStarVocalic
];
```

The prefix of *ġelīefan* supplies the upper boundary for **ga-* > **ge-*. If SC035 OEPrefixAReduction follows SC043 AngloFrisianBrightening, PGmc **galáubijaną* yields *ġealīefan* rather than expected OE *ġelīefan* 'believe'. Earlier placement changes no checked output, so the witness dates prefix reduction before brightening without locating its beginning.

26.3 Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell's discussion of *weorold* / *weoruld* and Ringe and Taylor's derivation of **weraldu* > **weruldu* > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

26.4 SC036. Raising of medial **a* between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising [
    {*a} -> {*u}
        || PGmcStarVowel EnglishStarConsonant* _
           [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
    {*à} -> {*u}
];
```

The two boundaries have unequal force. Before SC019 NWGmcFinalLongORaising, PGmc **sáiwalō* yields *sāwel* rather than expected OE *sāwol* 'soul'; after SC040 OEMedUnstressedULowering, it yields *sāwul* rather than *sāwol*, while PGmc **wír-àldu* yields *weoruld* rather than *weorold* 'world'. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In handbook terms, medial **a* > **u* belongs to the *world-* and *soul-*type low-stress vocalism that followed the earlier final-vowel changes.

26.5 Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as *reġnboga* 'rainbow'. Its only checked boundary is the immediately following technical stress-stripping stage, which is not a sound change.

26.6 SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope [
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _
           OEAnyConsonant+ PGmcStarGraveVowel
];
```

The *reġnboga* test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc **régna-bùgô* yields *reġnefoga* rather than expected OE *reġnboga* 'rainbow', because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

27 Medial unstressed vowel changes

27.1 Historical discussion of medial unstressed vowel changes

The history of *wuduwe* 'widow' orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* / *weoruld* alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu*, and **jugunþi* (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

27.2 SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

```
define OEWICombinativeUUmlaut [
    {*í} -> {*ú}
        || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

The *wuduwe* 'widow' derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut follows SC040 OEMedUnstressedULowering, PGmc **wíduwōn* yields *wudowe* rather than expected OE *wuduwe*; earlier placement changes no checked output. The witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

27.3 SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering [
    {*u} -> {*o}
        || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _
           [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

The two witnesses date medial unstressed **u* > **o* at very different scales. Before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* yields *wudowe* rather than expected OE *wuduwe* 'widow'; after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* yields *ġeogoþ* rather than expected *ġeoguþ* 'youth'. The local *weorold* 'world' and *widow* evidence places lowering after combinative u-umlaut, while the youth form supplies only the distant requirement that lowering precede unstressed long-vowel shortening.

28 Final bare-*a* loss

28.1 Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

28.2 SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

```
define PWGmcFinalBareALoss [
    {*a} -> 0 || _ .#.
];
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final *z*-deletion, the change gives the wrong outputs: PGmc **bárdaz* yields *bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* yields *camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* yields *craft* rather than expected OE *cræft* ‘craft’, and PGmc **dágaz* yields *dag* rather than expected *dæġ* ‘day’. The distant lower limit follows final *z*-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

29 Surviving bimoric **ō* unrounding

29.1 Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* 'rest' nevertheless undergoes unrounding before SC043 AngloFrisianBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90).

The sole witness establishes a local relation to brightening but supports no broader generalization.

29.2 SC042. Unrounding of the surviving bimoric **ō* (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding [
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
];
```

The single *ræste* 'rest' derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 PGmcFinalZDeletion or after SC043 AngloFrisianBrightening, PGmc **rástōz* yields *rasta* rather than expected OE *ræste*. Unrounding must therefore follow final *z*-loss and precede brightening, although only the relation to brightening is local.

30 Anglo-Frisian brightening

30.1 Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (Campbell 1959, 52, §131; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

30.2 SC043. Fronting of low **a* outside nasal environments (AngloFrisianBrightening)

```
define AngloFrisianBrightening [
    AngloFrisianBrighteningUnstressed .o.
    AngloFrisianBrighteningStressed .o.
    AngloFrisianBrighteningLongFinal
];
```

Two derivations place low **a* > **æ* between unrounding and breaking. Before SC042 PWGmc-SurvivingBimoricOUnrounding, PGmc **rástōz* yields *rasta* rather than expected OE *ræste* 'rest'. After SC044 OEBreaking, PGmc **sláxaną* yields *sleaan* | *slēaan* rather than expected OE *slēan* 'slay'. The first witness requires brightening to receive the outcome of the surviving-bimoric **ō* development; the second requires breaking to receive the fronted vowel.

31 Breaking and velar-fricative palatalization

31.1 Historical discussion of breaking and velar-fricative palatalization

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

Breaking has the fuller handbook treatment, while velar-fricative palatalization follows it locally in the *feoh* and *feohtan* type derivations.

31.2 SC044. Breaking before *h*, *rC*, and *lC* (`OEBreaking`)

```
define OEBreaking OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI;
```

Breaking must encounter the vowel created by brightening and must precede the fricative change seen in *feoh* 'fee' and *feohtan* 'fight'. Before SC043 AngloFrisianBrightening, PGmc **sláxaną* yields *sleaan* | *slēaan* rather than expected OE *slēan* 'slay'. After SC045 OEVelarFricativePalatalization, PGmc **féxu* yields *fehu* rather than expected OE *feoh*, and PGmc **féxtaną* yields *fehtan* rather than expected *feohtan*. The two feeding relations place breaking between brightening and velar-fricative palatalization.

31.3 SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization [
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*ɣ} -> {*j} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*ɣ} -> {*j} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j},
    {*ɣ} -> {*j} || _ {*j}
]
    .o. EnglishStarAlphabet*;
```

The local chronology comes from *feoh* and *feohtan*. Before SC044 OEBreaking, palatalization of **x* and **ɣ* beside front vowels or **j* makes PGmc **féxu* yield *fehu* rather than expected OE *feoh*, and PGmc **féxtaną* yield *fehtan* rather than expected *feohtan*. The distant upper limit comes from *six*: after SC060 OEWsPalatalUmlaut, PGmc **séxs* yields *sihs* rather than expected OE *six*. Breaking therefore feeds velar-fricative palatalization directly, while palatal umlaut supplies only the broader upper limit.

32 A-restoration and nasal changes

32.1 Historical discussion of A-restoration

Campbell's restoration of *a* before following back vowels and Ringe and Taylor's later retraction describe the same post-brightening development (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

32.2 SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration (
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 AngloFrisianBrightening, PGmc **bákaną* yields *bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* yields *færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákaną* again yields *bæcan* instead of *bacan*, while PGmc **wádaną* yields *wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

32.3 Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

32.4 SC047. Heavy-syllable nasal apocope of final **ą* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope [
    {*ą} -> 0 || OEAnyConsonant _ .#.
];
```

The evidence for final nasalized **ą* loss is sharply asymmetric. Before SC034 OEAwLongDiphthong, the single PGmc witness **stráwą* yields *stræw* rather than expected OE *strēaw* 'straw'. After SC048 OESecondaryNasalization, PGmc **bákaną* yields *bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* yields *binden* rather than expected *bindan* 'bind', alongside a broad *-en* failure set. One lower witness places apocope after long-diphthong formation; many reciprocal upper failures place it before secondary nasalization.

32.5 SC048. Secondary nasalization before final **n* (OESecondaryNasalization)

```
define OESecondaryNasalization [
    {*a} -> {*ą} || _ {*n} .#.
];
```

The broad *-an*/*-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* yields *bacen* rather than expected OE *bacan*, and PGmc **bíndaną* yields *binden* rather than expected *bindan*. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* yields *steolan* rather than expected OE *stelan* 'steal', and PGmc **wébaną* yields *weofan* rather than expected *wefan* 'weave'. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

33 B allophony and Sievers-law syncope

33.1 Historical discussion of B allophony

The first change is the positional alternation of Germanic **b*. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

The distribution is narrow, but later changes presuppose the stop-fricative alternation.

33.2 SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reġnboga* 'rainbow' develops as *reġnfoga* rather than expected OE *reġnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

33.3 Historical discussion of Sievers-law syncope

Sievers' Law concerns a different historical problem. It is a prosodic and morphological adjustment in heavy stems, not a distributional allophone of a stop consonant. Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

33.4 SC050. Sievers-law syncope (`SieversLawSyncope`)

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

The Sievers-law reduction **-CijV-** > **-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* yields *strecċan* rather than expected OE *streċċan* 'stretch'; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

34 Palatalization of **sk* to **sc*

34.1 Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992, 106–7, 111–12). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

34.2 SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

The non-fronted vowels of *flasce* 'flask' and *wascan* 'wash' fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* 'flask' and *wæscan* 'wash' rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sċeaft* 'shaft', *sċēar* 'shear', *sċēaþ* 'sheath', *sċēap* 'sheep', and *sċield* 'shield' before SC056 OEWsPalatalDiphthongization. The **sċea-*/*sċie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

35 Velar palatalization before front vowels

35.1 Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading “Frühe Verschiebungen in palataler Richtung,” he treats English k and g before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* ‘town’, *geaf* ‘gave’, *giefan* ‘give’, and *giest* ‘guest’, already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914--1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone sk cluster. His remark that “[sk] is more prone to palatalization and assibilation than [k]” is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* ‘child’, *dæg* ‘day’, *giefan* ‘give’, and *giest* ‘guest’, which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant j (Hogg 1992, 103–4). This formulation replaces a broad list of palatal outcomes with a phonological environment defined by adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that “after initial velars and **sk* had been palatalized” West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* ‘wake’, *licgan* ‘lie’, *lecgan* ‘lay’, *secg* ‘retainer’, *ecg* ‘edge’, *wicg* ‘horse’, and *brycg* ‘bridge’, illustrate the same point in lexical detail: front vowels and j create the palatal environment in which plain k and g cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Luick describes a broad early movement; Campbell distinguishes plain velars from the sk complex; Hogg specifies the adjacency and syllable conditions; and Ringe and Taylor order the plain-velar change before West-Saxon diphthongization. Plain-velar palatalization thus forms part of a wider palatalizing environment without being identical to its neighboring changes.

35.2 SC052. Palatalization of **k* before front vowels and **j* (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*ī́},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*ī́} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*ī́} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;
```

The *weccan* ‘wake’, *licgan* ‘lie’, and *lecgan* ‘lay’ set identifies front vowels and j as the environment for palatalization of k (Ringe and Taylor 2014, 213–14, §6.4.1). These forms establish the conditioning; different witnesses establish the chronology.

Applied before Sievers-law syncope, PGmc **strákkijaną* yields *strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after i-umlaut fronting, PGmc **kūi* and **lúnganjō* yield *ċȳ* ‘cows’ and *lunġen* ‘lungs’ rather than expected OE *cȳ* and *lungen*. The front-vowel k change therefore follows Sievers-law syncope and precedes i-umlaut fronting.

35.3 SC052. Velar palatalization before front vowels (OEVelarPalatalization)

```
define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

Plain k and g palatalization in front-vocalic and j-adjacent environments follows sk-palatalization and occupies a sharply defined pre-umlaut interval. Applied before Sievers-law syncope, PGmc **strákkijaną* yields *strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after general i-umlaut, PGmc **kūi* yields *ċȳ* rather than expected *cȳ* ‘cows’, and PGmc **lúnganjō* yields *lunġen* rather than expected *lungen* ‘lungs’. These witnesses place velar palatalization after Sievers-law syncope and before umlaut.

Luick, Campbell, and Ringe and Taylor place *cild* ‘child’ and *dæg* ‘day’ in a consonantal palatalization that precedes later vowel fronting (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1). The umlautal developments therefore receive plain k and g already reshaped beside front vowels and j.

The sk change belongs to the same palatalizing region with a separate scope. The *streċċan* ‘stretch’ evidence establishes a specific dependency on earlier syncope; it does not merge the two changes into one process.

36 Post-velar *w-loss and loss of *w before final *i

36.1 Historical discussion of early *w-loss before umlaut

The first rule is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor derive PGmc **singwan* to Old English *singan* 'sing' (Ringe and Taylor 2014, 214, §6.4.2). This comparative evidence establishes the change, although no checked form fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* 'sea' from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The first rule is restricted to the **ngw* sequence; the second has a specific lexical witness and defined earlier and later limits.

36.2 SC053. Loss of *w after velars (`OEPostVelarWLoss`)

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

The comparative development `*singwan > singan` establishes narrow post-velar **w*-loss in the **ngw* sequence, yielding *singan* 'sing'. Moving SC053 OEPostVelarWLoss earlier or later leaves every checked output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

36.3 SC054. Loss of *w before final *i (`OEWLossBeforeI`)

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

The history of *sǣ* 'sea' explains why non-initial **w* disappeared before final unstressed **i*. Campbell describes the loss, Ringe and Taylor derive the form from **saiwi-/*sawi-*, and Luick gives the parallel trajectory (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 PGmcFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields *sǣw* 'sea' rather than expected OE *sǣ*. The loss must therefore follow final *z*-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

37 The Old English i-umlaut and West Saxon palatal diphthongization

37.1 Historical discussion of i-umlaut and West Saxon palatal diphthongization

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* 'guest', *giefan* 'give', *hierde* 'shepherd', and *ieldra* 'older' (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992, 112). His examples, such as *bryd* 'bride', *trymman* 'strengthen', *bedd* 'bed', *ciest* 'chest', and *wiersa* 'worse', likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13). Ringe and Taylor's examples such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

37.2 SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* 'follow', *gylden* 'golden', *wyrm* 'worm', and *giest* 'guest' exemplify the same `i`- or `j`-conditioned fronting across different vowels (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The cow and lung forms establish the lower boundary. If fronting precedes velar palatalization, PGmc **kūi* yields *ċȳ* 'cows' rather than expected OE *cȳ*, and **lúnganjō* yields *lunġen* 'lungs' rather than expected OE *lungen*. The consonantal change must therefore precede fronting.

The gift and sheath forms establish the upper boundary. If West Saxon palatal diphthongization precedes fronting, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. Fronting consequently follows velar palatalization and precedes the West Saxon change; the other components of i-umlaut share those bounds.

37.3 SC055. Raising under i-umlaut (`OEIUmlautRaising`)

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted `æ` to `e` continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The same four forms fix both boundaries. If raising precedes velar palatalization, **kūi* yields *ċȳ* instead of expected *cȳ* and **lúnganjō* yields *lunġen* instead of expected *lungen*. If West Saxon palatal diphthongization precedes raising, **géftiz* yields *ġieft* rather than expected *ġift*, and **skáiθiz* yields *sċǣþ* rather than expected *sċēaþ*. These forms place raising after velar palatalization and before West Saxon palatal diphthongization.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before `m` and `n` in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

37.4 SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon `ie` forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

The chronology comes from the cow/lung and gift/sheath contrasts. Placed before velar palatalization, diphthongal mutation over-palatalizes **kūi* and **lúnganjō*; placed after West Saxon palatal diphthongization, it yields *ġieft* and *sċǣþ* instead of expected *ġift* and *sċēaþ*. These failures place diphthongal mutation after velar palatalization and before West Saxon palatal diphthongization.

37.5 SC055. The composite i-umlaut rule (`OEIUmlaut`)

```
define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong;
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The lower boundary is consonantal. If general umlaut precedes velar palatalization, PGmc **kūi* yields *ċȳ* 'cows' rather than expected *cȳ*, and PGmc **lúnganjō* yields *lunġen* 'lungs' rather than expected *lungen*. These over-palatalized forms place general umlaut after velar palatalization.

The upper boundary separates general umlaut from the narrower West Saxon process. If West Saxon palatal diphthongization precedes umlaut, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. Together the two witness pairs place general umlaut after velar palatalization and before the West Saxon process.

37.6 SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.]
];
```

West Saxon *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show diphthongization after an already palatal consonant (Ringe and Taylor 2014, 215–16, §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg's *giefan* 'give' and *sceap* 'sheep' belong to the same palatal-consonant environment (Hogg 1992, 108–9). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (Fulk 2018, 74, §4.13).

The forms *ġift* 'gift' and *sċēaþ* 'sheath' fix the lower boundary. If West Saxon palatal diphthongization precedes general i-umlaut, PGmc **géftiz* yields *ġieft* 'gift' rather than expected *ġift*, and PGmc **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. These witnesses place West Saxon palatal diphthongization after general umlaut; no tested lexical item supplies a later terminus ante quem.

The one-sided chronology reflects the difference in scale. General umlaut reorganizes the vowel system, whereas West Saxon palatal diphthongization affects a narrower dialectal class after palatal consonants. Its exact later placement remains undemonstrated by the present lexicon.

38 J-cluster coalescence

38.1 Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīeġan* 'bend' and *sēċan* 'seek' without assigning this later cluster adjustment the status of a major sound law (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

38.2 SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ʤ}]
    .o. [{*k} {*j} -> {*ʧ}]
);
```

The forms *bīeġan* 'bend' and *sēċan* 'seek' determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc `*báugijaną` yields *bēaġan* 'bend' rather than expected OE *bīeġan*, and PGmc `*sōkijaną` yields *sōċan* 'seek' rather than expected *sēċan*. This demonstrates that velar palatalization preceded coalescence. Nothing in the present lexicon supplies a terminus ante quem.

39 Nasal dissimilation

39.1 Historical discussion

Most accounts introduce nasal dissimilation to explain individual forms rather than as a regular sound law. Luick records *enetre* 'yearling' (spelled *enitre* in his text) (Luick 1914--1940, 166); Campbell discusses *heofon* 'heaven' with suffixal variation (Campbell 1959, 155); and Hogg encounters the same form while treating back mutation (Hogg 1992, 112).

Fulk supplies the clearest general formulation: "In the cluster mn, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular" (Fulk 2018, 121, §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* 'yearling' reflects "loss of the second **n* by dissimilation" (Ringe and Taylor 2014, 282).

The disagreement concerns scope. Fulk's formulation recognizes a recurrent but irregular development in `mn`; the remaining discussions stay with particular lexical outcomes. None warrants a sound law comparable in scope to the major Old English vowel changes.

39.2 SC058. Nasal dissimilation in short-vowel environments (OENasalDissimilation)

```
define OENasalDissimilation [
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n}
     ↪  [EnglishStarShortVowel | .#.]
];
```

I adopt a narrower environment than the handbook observations might suggest. Fulk formulates the tendency at the level of `mn` clusters and illustrates it with *heofon* 'heaven' and *fæstenn* 'fasting' (Fulk 2018, 121, §6.11). Ringe and Taylor show the same kind of development in *enetre* 'yearling' (Ringe and Taylor 2014, 282). Campbell's "*heofon* is for older *hefzen*" and Hogg's sequence **hefon* > *heofon* preserve outcomes of the same kind (Campbell 1959, 155; Hogg 1992, 112). The short-vowel environment adopted here covers a recurrent subset of these outcomes, not every dissimilatory development involving nasals.

No witness word fixes the position of nasal dissimilation within the Old English sequence. Reversing its order with any tested neighbor leaves every checked output unchanged. A more precise relative chronology would therefore require lexical evidence not represented here.

40 Back mutation

40.1 Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* and *weofan*. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (Ringe and Taylor 2014, 319, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (Campbell 1959, 86, §207), while Hogg draws the instructive comparison with breaking (Hogg 1992, 112). Fulk accordingly separates back mutation from the earlier umlautal changes (Fulk 2018, 69, §4.8).

40.2 SC059. Back mutation before labials and liquids (`OEBackMutation`)

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébaną* produce *ġeofan* ‘give’; the expected form is *ġiefan* ‘give’. **stélaną* likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébaną* yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

41 West Saxon palatal umlaut

41.1 Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

41.2 SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage *mieht* and *nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

42 Weak-tail nasal loss

42.1 Historical discussion

The pathway from **dōną* to *dōn* 'do' supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120–21; Fulk 2018, 91, §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

42.2 SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

Final weak-tail **-ną* and **-mą* accordingly yield plain **-n* and **-m*.

Only *dōn* 'do' constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* 'do'. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

43 High-vowel apocope

43.1 Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

43.2 SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
```

```
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    {*u} -> 0 || {*x} _ .#.,
    {*ų} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
];
```

Final **i*, **u*, and **ų* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* yields *cū* rather than expected OE *cȳ* ‘cow’, and PGmc **brūdiz* yields *brūd* rather than expected OE *brȳd* ‘bride’. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* yields *fyrht* rather than expected OE *fyrhte* ‘fright’. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

44 Post-apocope **n*-loss and medial syncope

44.1 Historical discussion of post-apocope **n*-loss and medial syncope

Evidence for post-apocope reduction is strikingly uneven. The inherited **furht-* family makes the survival of one nasal diagnostic and fixes both sides of stem-final n-loss (Kroonen 2013, 201). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

44.2 SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

Only final **n* after long **ī* is at issue, as in the inherited family behind *fyrhte* 'fright'.

The same proto-form fixes both edges. Before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* yields *fyrhten* rather than expected OE *fyrhte* 'fright'. After SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* again yields *fyrhten* rather than expected *fyrhte*. I therefore order final bare-a loss, stem-final n-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the derivation, but depend upon one lexical family.

44.3 SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

```
define OEMedialSyncope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
];
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every checked output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

45 Late syncope and degemination

45.1 Historical discussion of late syncope and degemination

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *netele* supplies a concrete lexical type (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Brunner 1965, 144–45, §§158–159). Fulk places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

45.2 SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope [
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátilōn* yields *nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* yields *spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields *spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

45.3 SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation [
    {*θ} -> 0 || {*t} _
];
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (Hogg 1992, 120–21; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

45.4 SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN;
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* 'spindle' (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* yields *spinnl* rather than expected OE *spinl* 'spindle'. Syncope must therefore create the cluster before degemination simplifies it. Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

46 Early o-shortening

46.1 Historical discussion

After the principal palatal and umlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

46.2 SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 NWGmcNStemNLoss, PGmc **nḗdrōn* yields *nǣdran* rather than expected OE *nǣdre* 'adder', PGmc **érθōn* yields *eorþan* rather than expected *eorþe* 'earth', and PGmc **fláskōn* yields *flascan* rather than expected *flasce* 'flask'. The same earlier shift also disrupts forms such as *heorte* 'heart' and *līne* 'line'. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 NWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore do not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

47 Early unstressed fronting and later o-shortening

47.1 Historical discussion of early unstressed fronting and later o-shortening

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no checked form fixes its later boundary.

47.2 SC070. Early fronting of unstressed **a* (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

The rule fronts unstressed **a* to **æ* after the earlier shortening has created a frontable vowel but before the later shortening of unstressed **ō*. It produces endings such as OE *-en* in *lungen* 'lungs'.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* yields *lunġen* rather than expected OE *lungen* 'lungs'. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* yields *boreþ* rather than expected OE *boraþ* 'bears', and PGmc **mḗnōθz* yields *mōneþ* rather than expected *mōnaþ* 'month'. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

47.3 SC071. Later shortening of unstressed **ō* (`OELateOShortening`)

The following rule handles the later shortening stage.

```
define OELateOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
];
```

The rule shortens the remaining unstressed long **ō* after fronting, producing the later "stable a" endings in OE *boraþ* 'bears' and *liornaþ* 'learns'.

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* yield *boreþ* rather than expected OE *boraþ*, and PGmc **líznōθi* yield *liorneþ* rather than expected *liornaþ*. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly. Moving it later within the tested range creates no equally sharp failure.

48 Unstressed long-vowel shortening and ae-merger

48.1 Historical discussion of unstressed long-vowel shortening and ae-merger

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **æ* with **e* (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

48.2 SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8;
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* yields *fyrhten* rather than expected OE *fyrhte* 'fright'. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nḗdrōn* yields *nǣdræ* rather than expected OE *nǣdre* 'adder', and PGmc **fádēr* yields *fædær* rather than expected *fæder* 'father'. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

48.3 SC073. Merger of unstressed **æ* with **e* (`OEUnstressedAEMerger`)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

The rule merges unstressed **æ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **nḗdrōn* yields *nǣdræ* rather than expected OE *nǣdre*, and PGmc **fádēr* yields *fædær* rather than expected *fæder*. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* yields *tāæ* rather than expected OE *tā* 'toe'. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The checked forms fix the local order after SC072 OEUnstressedLongVowelShortening and place the merger before the later h-loss and contraction.

49 Medial unstressed-i lowering

49.1 Historical discussion of medial unstressed-i lowering and **ng* retention

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (Hogg 1992, 120–21; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *sċilling* 'shilling' type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

49.2 SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 [
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _
];
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* yields *fyrhti* rather than expected OE *fyrhte* 'fright'. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* yields *sċilleng* rather than expected *sċilling* 'shilling'. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

49.3 SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering [
    {*e} -> {*i} || _ {*n} {*g}
];
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* 'shilling'.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skíllingaz* yield *sċilleng* rather than expected OE *sċilling*. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

50 Prefix i-reduction

50.1 Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk's discussion of prefix vowels accounts for OE **be-* and **ne-* (Fulk 2018, 97, §5.7). Hogg and Ringe and Taylor place such weakening within the broader late history of unstressed vowels (Hogg 1992, 120–21; Ringe and Taylor 2014, 298–332, §§6.8.3–6.9.6).

The tested forms do not determine the rule's position relative to a neighboring change.

50.2 SC076. Reduction of prefixal **i* in unstressed position (OEPrefixIReduction)

```
define OEPrefixIReduction [
    {*i} -> {*ĕ} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] EnglishStarVocalic
];
```

The rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. The development accounts for later prefix spellings such as OE **be-* and **ne-*.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC076 OEPrefixIReduction before or after any specific neighboring change.

The handbooks attest late prefix-vowel weakening, but the precise placement remains approximate. No lexical failure fixes it.

51 Weak-tail reduction

51.1 Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

51.2 SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc **bákaną* yields *bacen* rather than expected OE *bacan* 'bake', and PGmc **híndaną* yields *binden* rather than expected *bindan* 'bind', alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc **fléuxaną* yields *flēoan* rather than expected OE *flēon* 'flee', and PGmc **sláxaną* yields *sleaan* rather than expected *slēan* 'slay'.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

52 Final-j loss and final geminate simplification

52.1 Historical discussion of final-j loss and final geminate simplification

After SC079 OEJLossAfterHeavy removes **j* in heavy environments, forms such as *lungen* 'lungs' acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final *nn* outcome in the following derivation.

52.2 SC079. Loss of **j* after heavy syllables (`OEJLossAfterHeavy`)

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

The rule removes **j* after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is **j*.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* yields *ġelēafan* rather than expected OE *ġelīefan* 'believe', PGmc **báugijaną* yields *bēaġan* rather than expected *bīeġan* 'bow', and PGmc **fúlgijaną* yields *fulġan* rather than expected *fylġan* 'follow'. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* yields *lungenn* rather than expected OE *lungen* 'lungs'. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

52.3 SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* yield *lungenn* rather than expected OE *lungen*. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

53 J-strengthening, vocalization, and ei-contraction

53.1 Historical discussion of j-strengthening, vocalization, and ei-contraction

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then vocalizes the remaining intervocalic **j*, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

53.2 SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ḗa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjaną* yields *strēaġan* rather than expected OE *strīeġan* ‘strew’. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields *strīeian* rather than *strīeġan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīeġan* derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

53.3 SC082. Intervocalic vocalization of **j* (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule vocalizes intervocalic **j* to **i*, creating the *ei*-like sequence later removed by SC083 OEUnstressedEIContraction in many weak verb forms.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjaną* yield *strīeian* rather than expected OE *strīeġan* ‘strew’. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjaną* yield *boreian* rather than expected OE *borian* ‘bore’, PGmc **xándlōjaną* yield *handleian* rather than expected *handlian* ‘handle’, and PGmc **mákōjaną* yield *maceian* rather than expected *macian* ‘make’. The witness forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

53.4 SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
```

```
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ {*i}
];
```

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* ‘bore’ and *liccian* ‘lick’.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **búrōjanq̨* yield *boreian* rather than expected OE *borian*, PGmc **líznōjanq̨* yield *liorneian* rather than expected *liornian*, and PGmc **líkkōjanq̨* yield *licceian* rather than expected *liccian*. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

54 H-loss and contraction

54.1 Historical discussion of h-loss and contraction

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (Fulk 2018, 270, §12.21), and Luick describes the corresponding West Germanic contractions (Luick 1914--1940, 165).

54.2 SC085. Loss of intervocalic **h* (`OEHLoss`)

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* yields *tāæ* rather than expected OE *tā* 'toe'. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* yields *flēoan* rather than expected OE *flēon* 'flee', PGmc **sláxaną* yields *sleaan* rather than expected *slēan* 'slay', PGmc **téxun* yields *teoon* rather than expected *tēon* 'draw', and PGmc **táixōn* yields *tāe* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

54.3 SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*ḗo},
    {*ḗo} {*o} -> {*ḗo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*ḗ} {*a} -> {*ḗ},
    {*ḗ} {*e} -> {*ḗ},
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*ī́} {*a} -> {*ī́},
    {*ī́} {*e} -> {*ī́},
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
```

```
    {*ū} {*e} -> {*ū}
];
```

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* 'flee', *slēan* 'slay', and *tēon* 'draw'.

Moving contraction before SC085 OEHLoss makes PGmc **fléuxaną* yield *flēoan* rather than expected OE *flēon*, PGmc **sláxaną* yield *sleaan* rather than expected *slēan*, PGmc **téxun* yield *teoon* rather than expected *tēon*, and PGmc **táixōn* yield *tāe* rather than expected *tā*. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

55 R-metathesis

55.1 Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* 'burst', *forst* 'frost', and *cærse* 'cress' (Brunner 1965, 159, §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (Luick 1914--1940, 201).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

55.2 SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* 'burst' where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstaną* yield *beorstan* rather than expected OE *berstan* 'burst'. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters none of the checked outputs.

The checked forms fix the earlier relation but do not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

56 References

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